

1. Convert to while loop

```
for (int j = 10; j > 0; j -= 2) {  
    System.out.println(j + " ");  
}
```

2. convert to do while loop

```
int x = 0;  
while (x < 15) {  
    x += 3;  
}
```

3. Rewrite using a for loop and write the output

```
int a = 5;  
int b = 12;  
while (a <= b) {  
    System.out.println("a = " + a);  
    a += 2;  
    b--;  
}
```

4. Convert to a do-while loop and write the output

```
int num = 8;  
while (num > 0) {  
    System.out.println(num * 2);  
    num--;  
}
```

5. Identify and correct the error

```
int count = 1;  
while (count <= 10);  
{  
    System.out.println(count);  
    count++;  
}
```

6. Rewrite using a while loop and write the output

```
int m = 4;  
for (int n = 1; n <= 3; n++) {  
    m += n;  
}  
System.out.println("m = " + m);
```

7. Write the output and number of times the loop will be executed

```
int a = 2, b = 8;  
while (a < b) {  
    a *= 2;  
    b -= 3;  
    System.out.println("a = " + a + ", b = " + b);  
}
```

8. Write the output and number of times the loop will be executed

```
int k = 3;
for (int i = 1; i <= 4; i++) {
    k += i;
    System.out.println("k = " + k);
}
```

9. Find the errors if any and write the correct code

```
int ctr = 0;
while (ctr != 5)
ctr++;
{
    System.out.println("Counter: " + ctr);
}
```

10. Write the output and number of times the loop will be executed

```
int p = 50;
while (p >= 5) {
    p /= 2;
    System.out.println(p);
}
```

11. Write the output and the number of times the loop will be executed

```
int a = 2, b = 1;
while (b++ < 5) {
    a *= b;
}
System.out.println(a);
```

12. Write the output and the number of times the loop will be executed

```
int x = 10;
while (-x > 5) {
    System.out.print(x + " ");
}
System.out.print(x);
```

13. Write the output and the number of times the loop will be executed

```
int n = 3, m = 2;
while (n-- > 0) {
    m += n;
    System.out.print(m + " ");
}
System.out.println(n);
```

14. Write the output and the number of times the loop will be executed

```
int p = 4, q = 1;
while (q++ < p) {
    p--;
    System.out.print(p + " ");
}
System.out.println(q);
```

15. Write the output and the number of times the loop will be executed.

```
int a = 50;
while (true) {
    if (a <= 0) break;
    a -= 15;
}
System.out.println(a);
```

16. Write the output and the number of times the loop will be executed.

```
int n = 300;
while (true) {
    if (n <= 50) break;
    n -= 60;
}
System.out.println(n);
```

17. find output :

```
int a = 1;
for (int i = 1; i < 5; i++) {
    if (i == 3) continue;
    a *= i;
}
System.out.println(a);
```

18. Write the output and the number of times the loop will be executed

```
int p = 5, k = 1;
while (k++ < 4) {
    if (k == 2) continue;
    p -= k;
}
System.out.println(p);
```

18. find output

```
String word = "JAVA";
for (int i = 0; i < word.length(); i++) {
    for (int j = 0; j <= i; j++) {
        System.out.print(word.charAt(j));
    }
    System.out.println();
}
```

New Level Unlocked !!!!

19. find output:

```
int a = 1, b = 10;
for (int i = 1; i <= 4; i++) {
    while (b-- > 5) {
        if (b % 2 == 0) {
            continue;
        }
        a += b;
        if (a % 3 == 0) {
            break;
        }
    }
    a *= i;
}
System.out.println("a = " + a);
System.out.println("b = " + b);
```

20. find output

```
int x = 2, y = 8, z = 1;
for (int i = 1; i <= 3; i++) {
    while (x < y) {
        if (x % 2 != 0) {
            x += 3;
            continue;
        }
        z += x;
        x++;
        if (z % 4 == 0) {
            z /= 2;
            break;
        }
    }
    y -= i;
    x += y;
}
System.out.println("x = " + x);
System.out.println("y = " + y);
System.out.println("z = " + z);
```

Final Boss !!

```
String result = "";
int sum = 0;

outerLoop:
for (int i = 65; i < 70; i++) { // ASCII range from 'A' to 'E'
    int x = i, y = 0;

    while (x > 60) {
        if (x % 2 == 0) {
            x -= 3;
            continue outerLoop;
        }

        char ch = (char) x; // Convert ASCII value to character
        y = ch + i;
        result += Integer.toString(y).charAt(1);

        if (y % 5 == 0) {
            sum += y;
            x -= 2;
        } else {
            x -= 4;
        }
    }

    if (result.length() % 2 == 0) {
        continue outerLoop;
    }

    sum += i;
}

System.out.println("Result: " + result);
System.out.println("Sum: " + sum);
```